When the patrols leave, mining comes back for seconds.

Operation Mercurio, La Pampa, from February 2019 · what it means for FZS in Tambopata

TWO FACTS ABOUT OUR ANIMAL

35,000

ants a day, with a 60 cm tongue flicking up to 150 times a minute

1 minute

per mound. It never destroys the nest: it moves on and comes back later

Smithsonian's National Zoo; National Geographic

Photo: giant anteater at a termite mound

FZS pays for operations. What happens when one ends?

Net mining additions, ha/yr · Tambopata buffer zone, La Pampa, Madre de Dios · 2016–18 vs 2019–21 vs 2022–25

WHAT WE FOUND

1 · IT FELL, LOCALLY

1,640 → 163

ha/yr, Tambopata buffer zone. 25%–47% of the fall: pits going out of use

2 · IT MOVED

60 → 395

ha/yr just across the road, in the legal corridor

3 · IT CAME BACK

1,829

ha/yr in the later years (2022–25), much of it on old pits

All three: description. Net additions = new land + re-mined − stopped, from yearly 30-m satellite maps.

DESCRIPTION

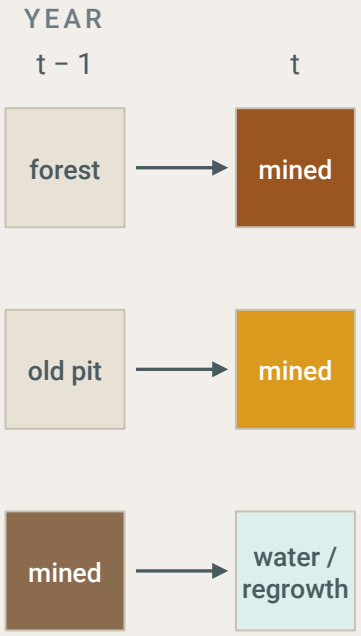
One measure on every slide: net mining additions

Net additions (ha/yr) = new land + re-mined − stopped · per 30-m pixel, year t vs the years before · MapBiomas Peru Collection 4, 1985–2025

+ New land
never mined before

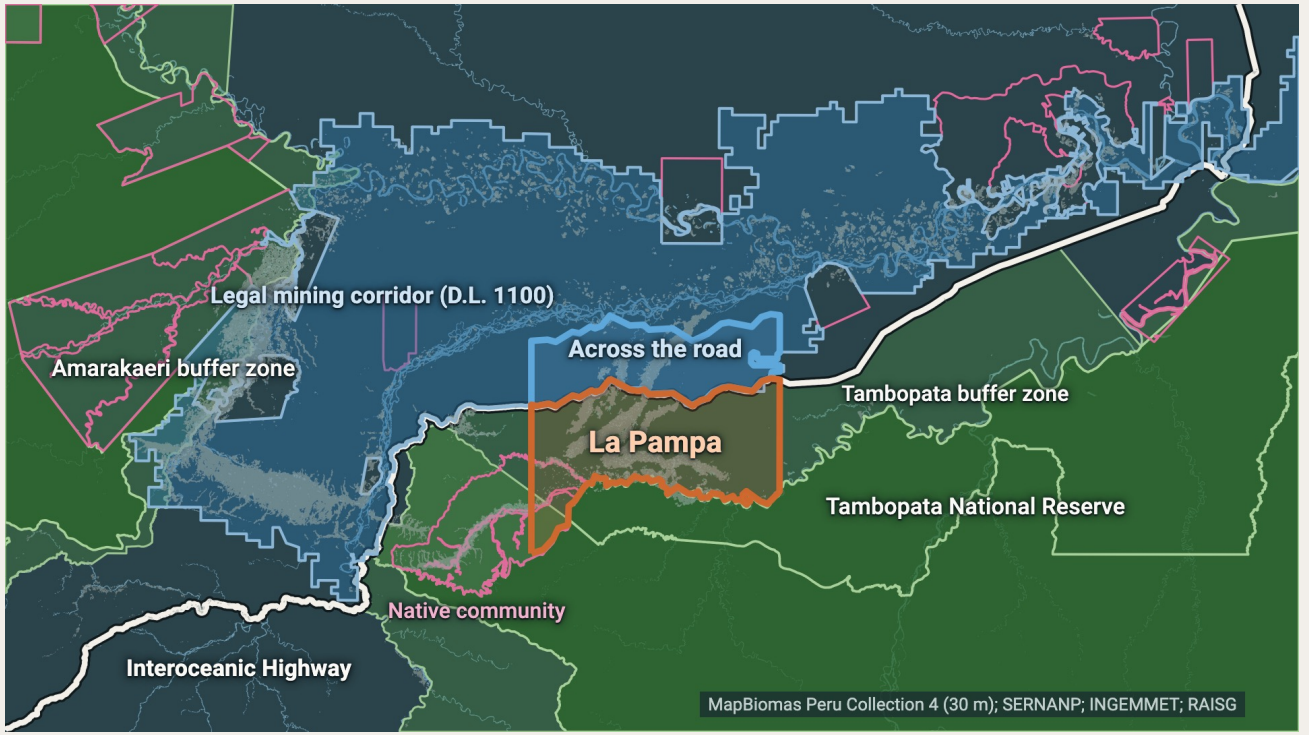
+ Re-mined
mined years ago, not last year

− Stopped
mined last year, not this year



Net = new + re-mined − stopped

= mining area this year − last year. Hectares are a proxy for mining activity, not for miners.



Same zones everywhere: Tambopata buffer zone (whole, incl. Kotsimba) · La Pampa sector (its 70.15–69.80° W part) · corridor across the road (D.L. 1100, ≤ 10 km north of the highway) · Amarakaeri buffer zone (less treated) · Madre de Dios.

DESCRIPTION

La Pampa sits one road away from legal mining land

Madre de Dios, Peru · step 1 of 4 · zones as used on every slide

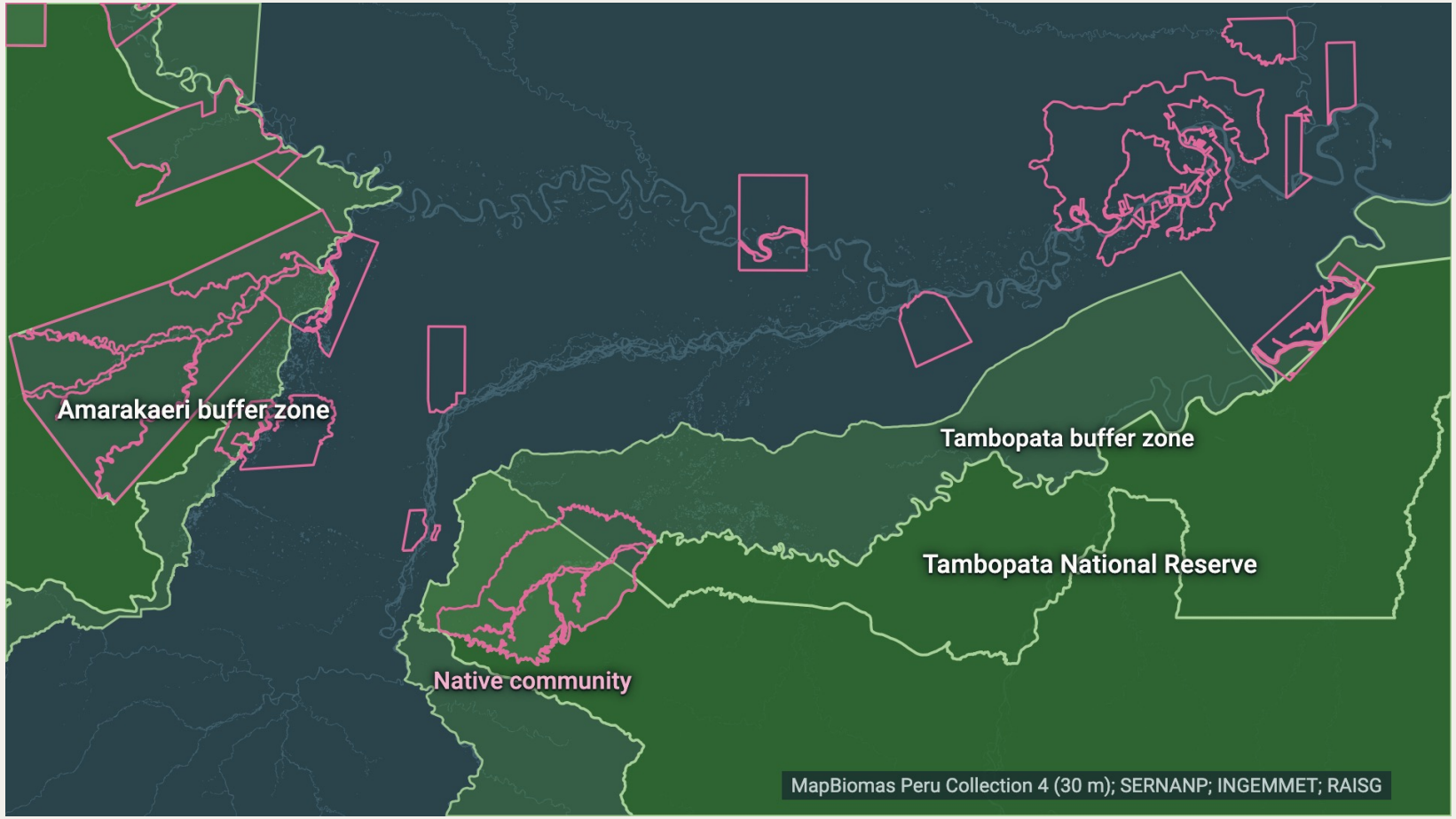
1 **Protected areas**

Tambopata reserve, its buffer zone and the Amarakaeri buffer zone. Pink: communities.

2 The Interoceanic Highway

3 Where mining can be legal

4 The target



DESCRIPTION

La Pampa sits one road away from legal mining land

Madre de Dios, Peru · step 2 of 4 · zones as used on every slide

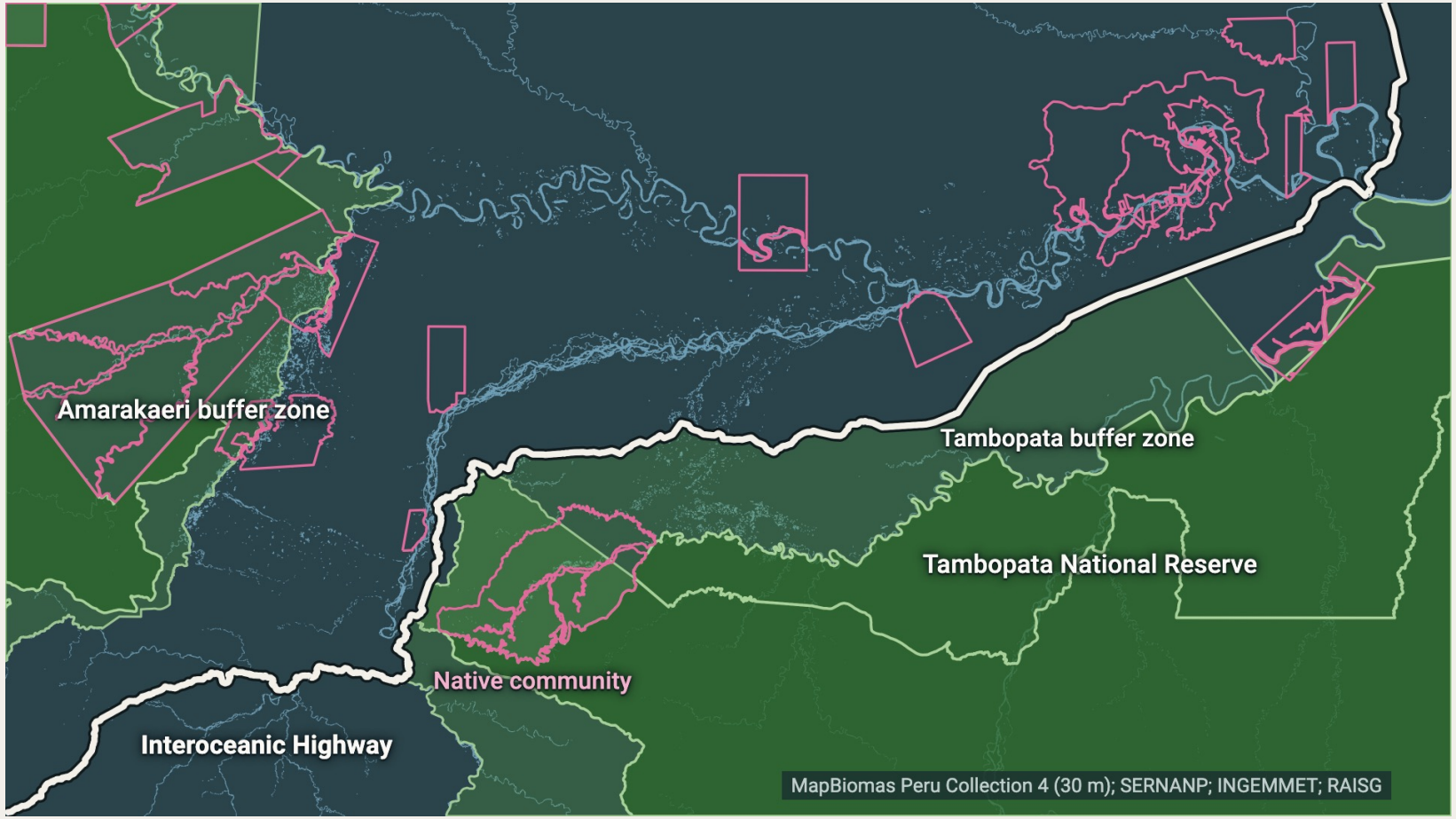
1 Protected areas

2 The Interoceanic Highway

One road runs along the buffer zone and splits it from the land to the north.

3 Where mining can be legal

4 The target



DESCRIPTION

La Pampa sits one road away from legal mining land

Madre de Dios, Peru · step 3 of 4 · zones as used on every slide

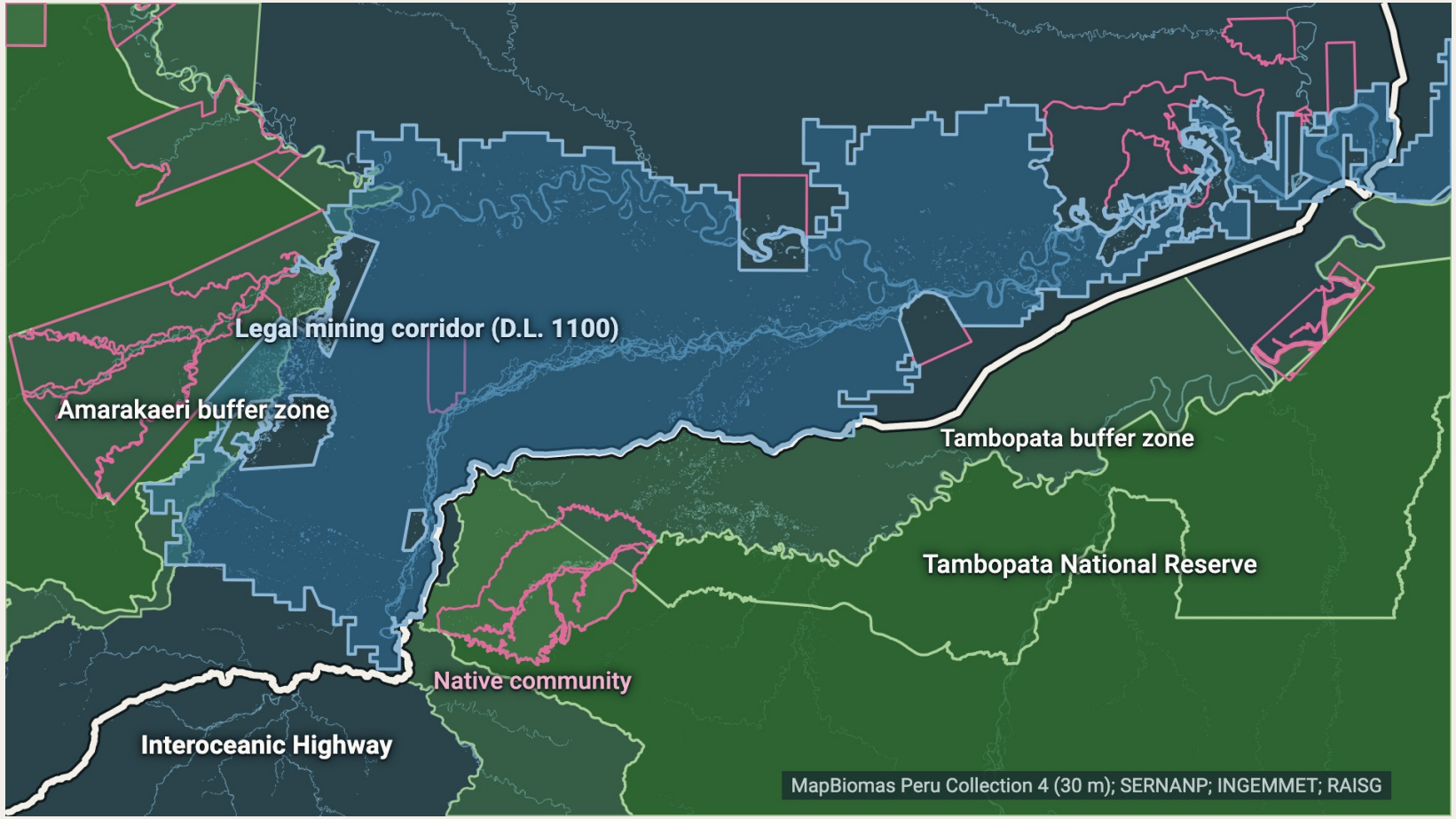
1 Protected areas

2 The Interoceanic Highway

3 Where mining can be legal

The legal mining corridor, north of the road. South of it, in the buffer zone: a crime.

4 The target



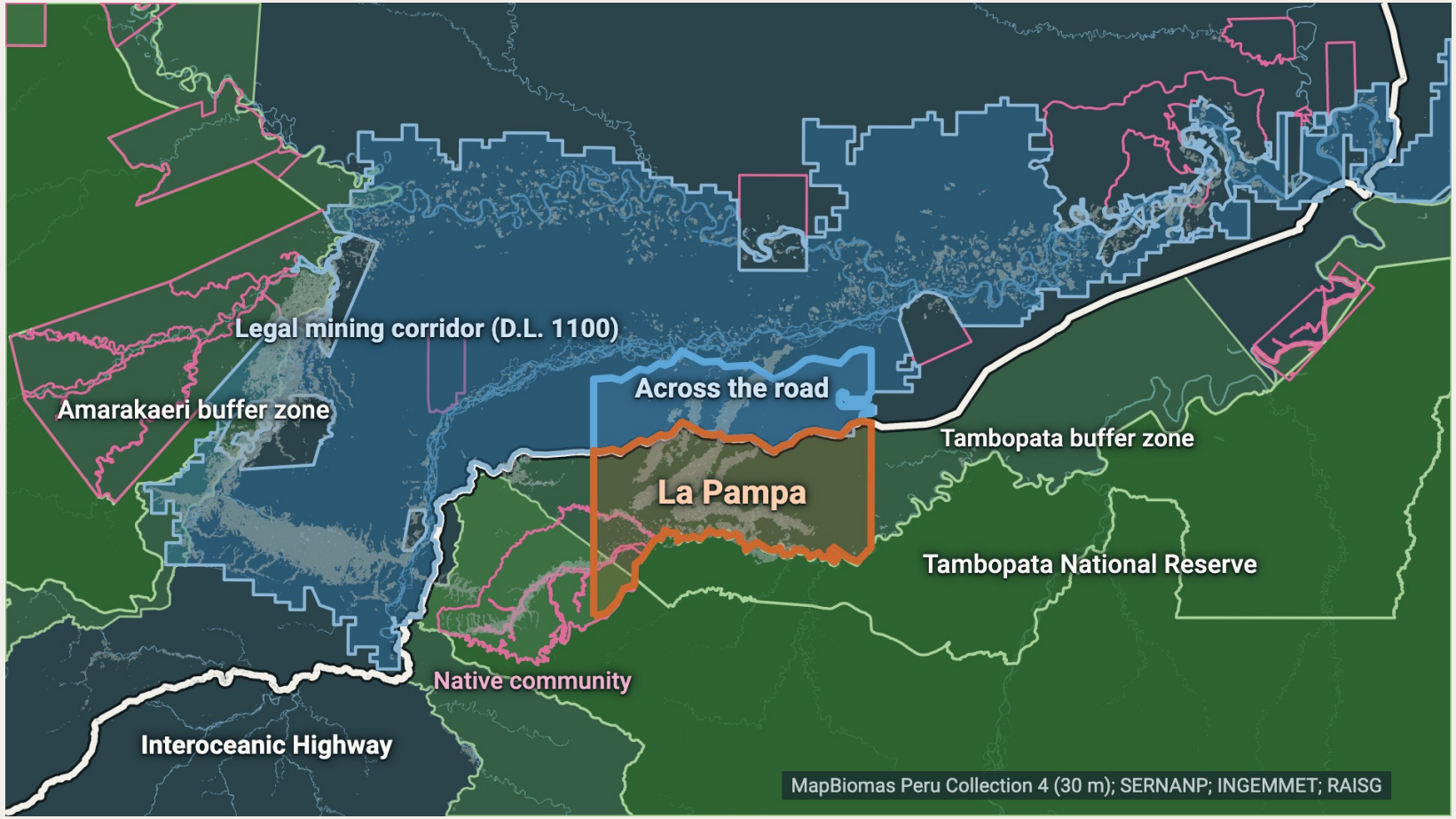
DESCRIPTION

La Pampa sits one road away from legal mining land

Madre de Dios, Peru · step 4 of 4 · zones as used on every slide

1. Protected areas
2. The Interoceanic Highway
3. Where mining can be legal
4. **The target**

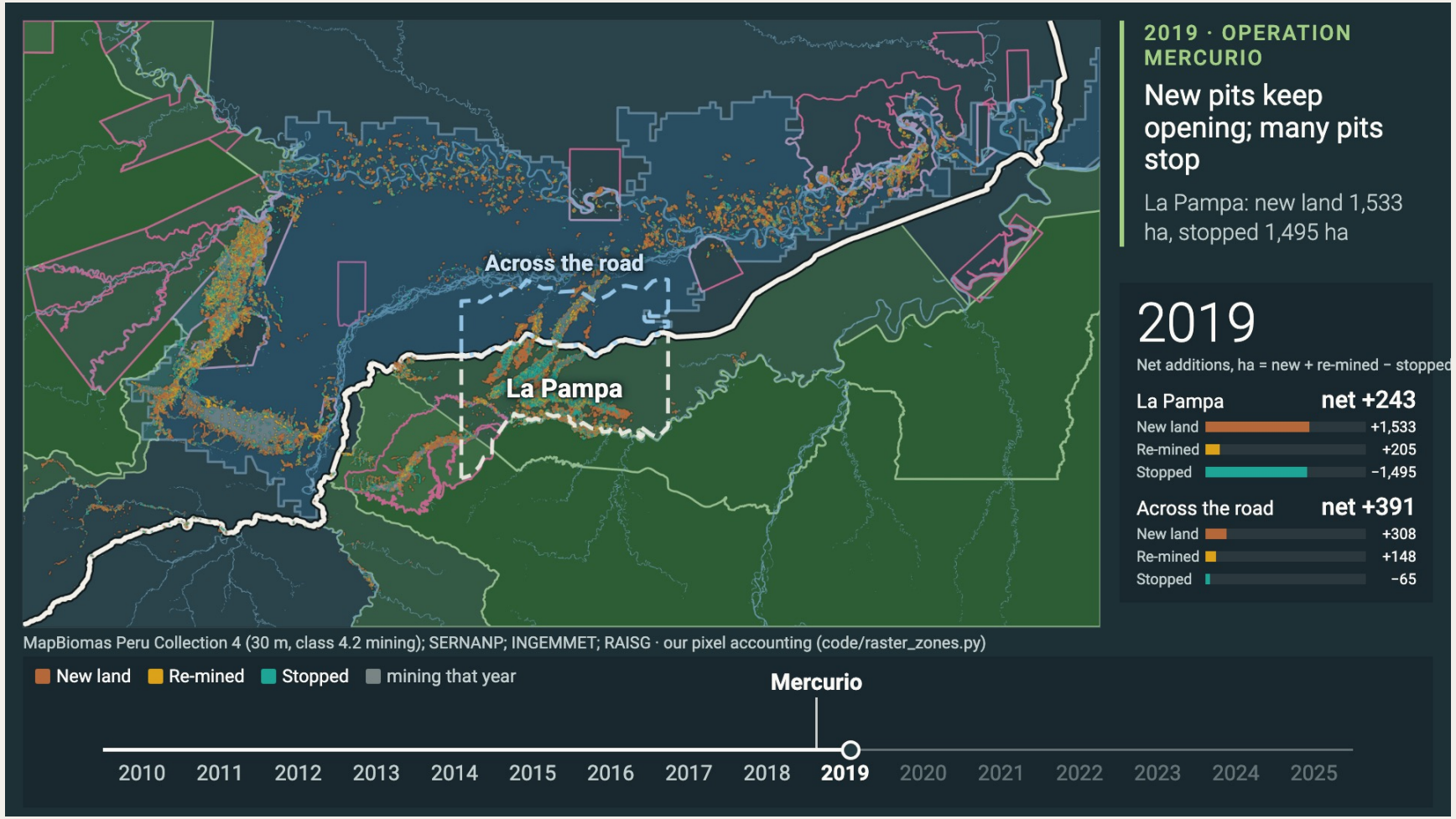
   La Pampa, Mercurio's target from Feb 2019, and the corridor strip across the road.



DESCRIPTION

2010–2025: new pits, old pits re-mined, pits stopping

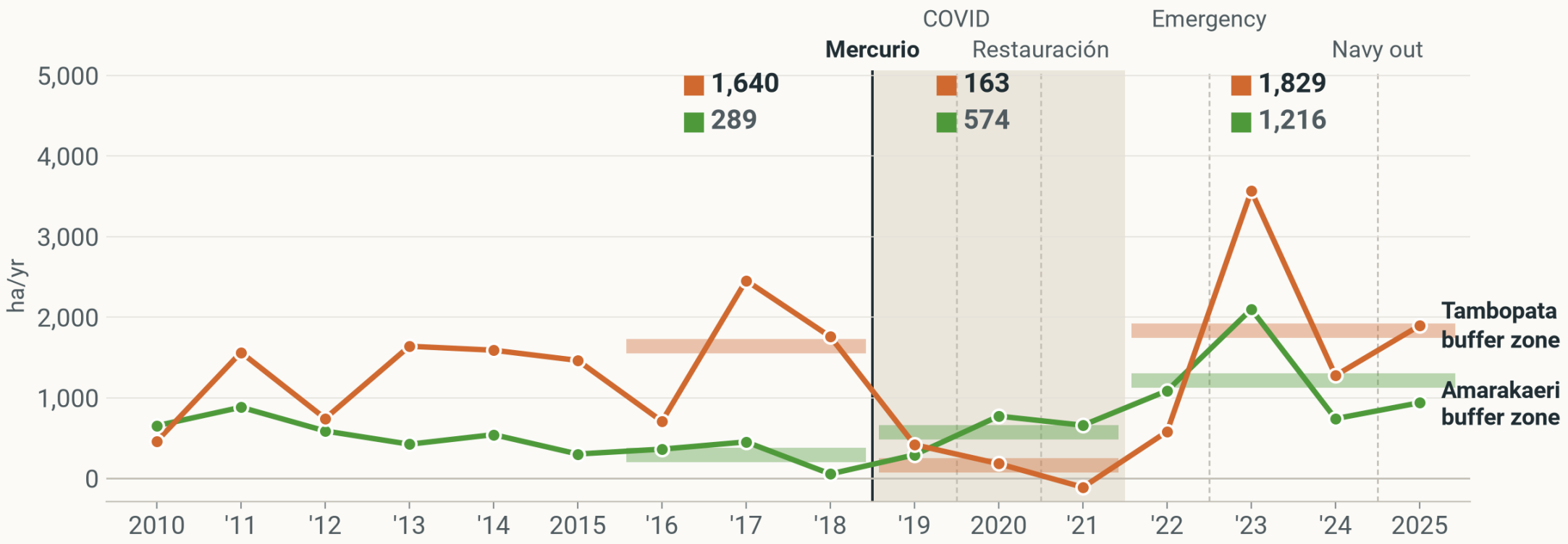
Net mining additions as three flows on the map: new land (brown), re-mined (amber), stopped (teal) · counter: net ha per year, La Pampa sector and corridor across the road



DESCRIPTION

Net additions in the Tambopata buffer zone fell 90% in 2019–21

Net mining additions, ha/yr · Tambopata buffer zone (orange) vs Amarakaeri buffer zone (green) · 2010–2025 · Tambopata 1,640 → 163, Amarakaeri 289 → 574 (2016–18 → 2019–21). Thick bars and numbers at the top: period means.

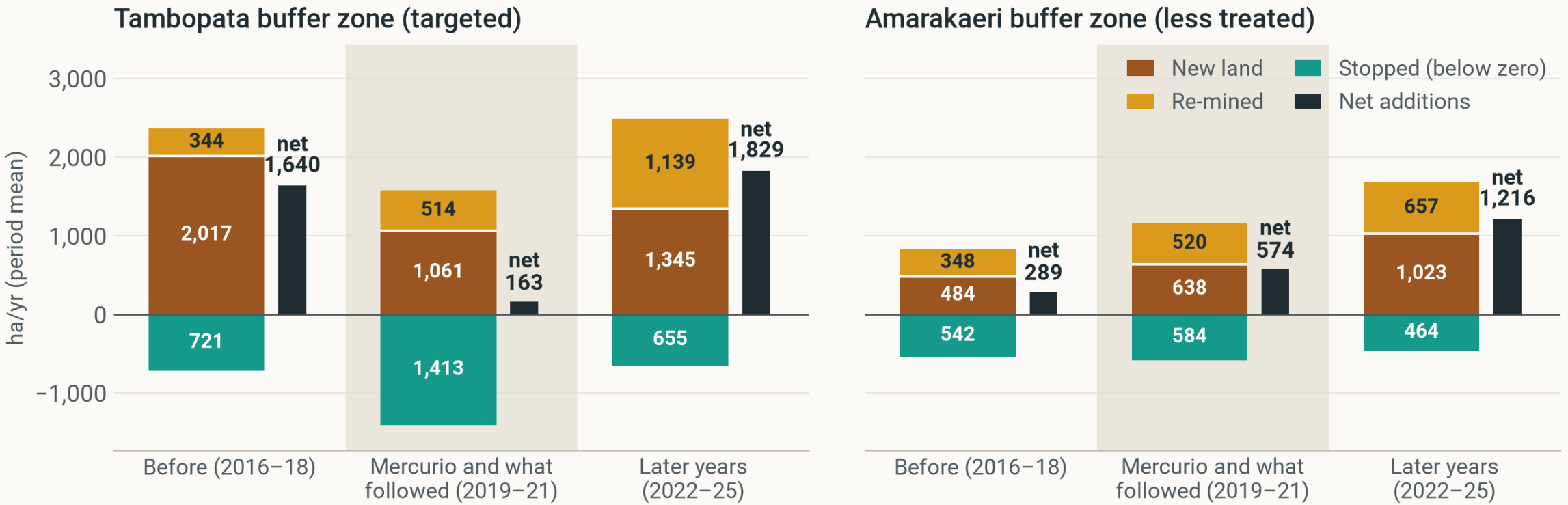


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), our pixel accounting on one zone registry (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; corridor = D.L. 1100). Both buffer zones whole (official SERNANP polygons). Satellites cannot see river dredges or mercury.

DESCRIPTION

Pits going out of use made 25%–47% of the fall, most of it in 2019

Net mining additions and their parts, ha/yr, period means · Tambopata buffer zone vs Amarakaeri buffer zone · Tambopata 2016–18 → 2019–21: new land 2,017 → 1,061, stopped 721 → 1,413 (2.0×). Stopped land makes 25%–47% of the fall (four ways of counting map flicker); 68% of the extra stopping is in 2019.

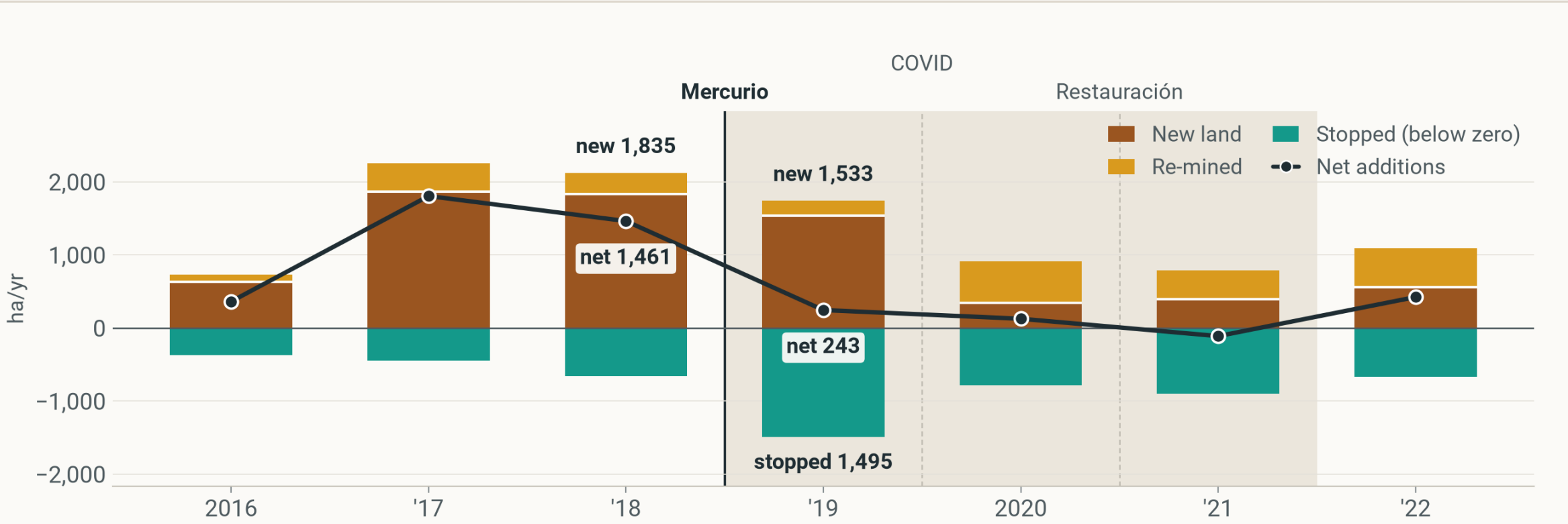


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), our pixel accounting on one zone registry (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; corridor = D.L. 1100). New land = mined for the first time since 1985; re-mined = mined again after a gap; stopped = left the mining class. Net = new + re-mined − stopped. Satellites cannot see river dredges or mercury.

DESCRIPTION

New pits kept opening in 2019; pits stopped where the patrols were

Net mining additions and their parts, ha/yr · La Pampa sector (the target) · 2016–2022 · 2019: new land 1,533 (2018: 1,835), stopped 1,495, net 243. Shaded: 2019–21.

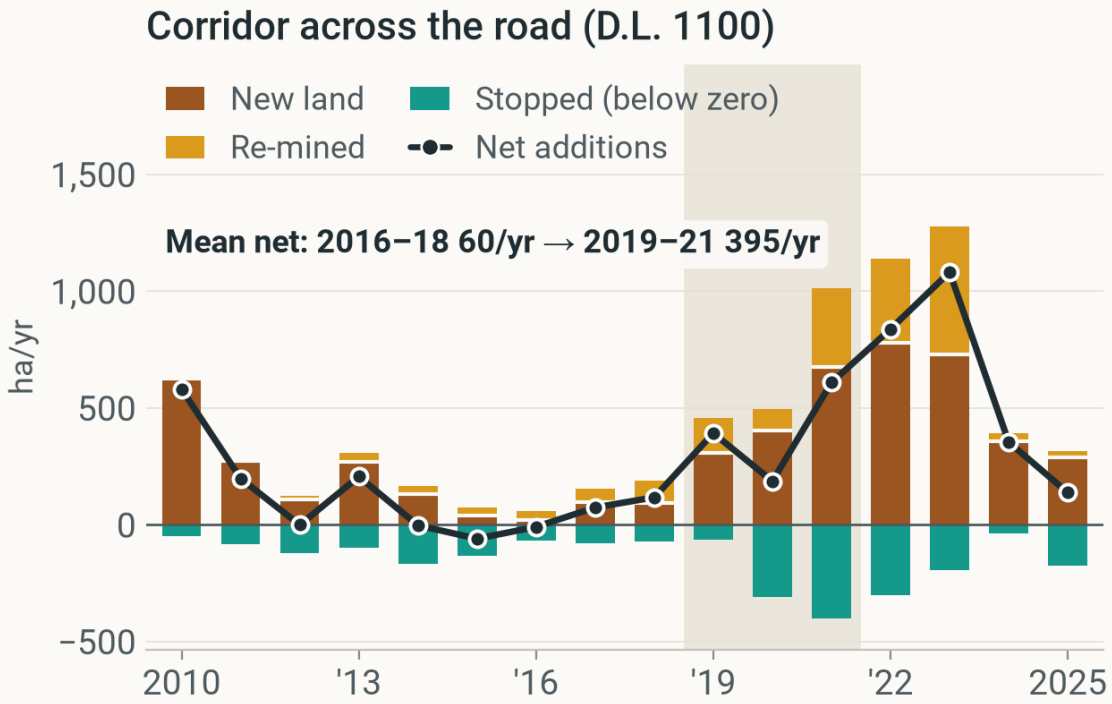


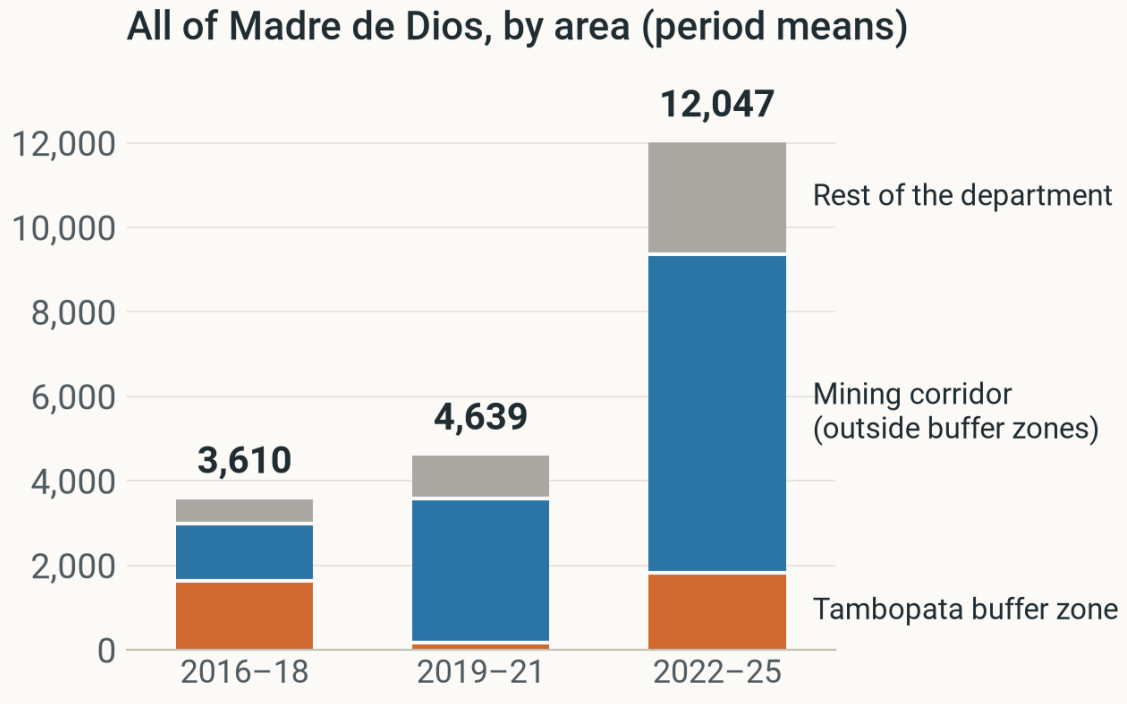
Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), our pixel accounting on one zone registry (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; corridor = D.L. 1100). New land = mined for the first time since 1985; re-mined = mined again after a gap; stopped = left the mining class. Satellites cannot see river dredges or mercury.

DESCRIPTION

Mining moved across the road; the department never fell

Net mining additions (and parts), ha/yr · left: corridor across the road, 2010–2025, 60 → 395 (2016–18 → 2019–21) · right: Madre de Dios 3,610 → 4,639 → 12,047 (2016–18 → 2019–21 → 2022–25).



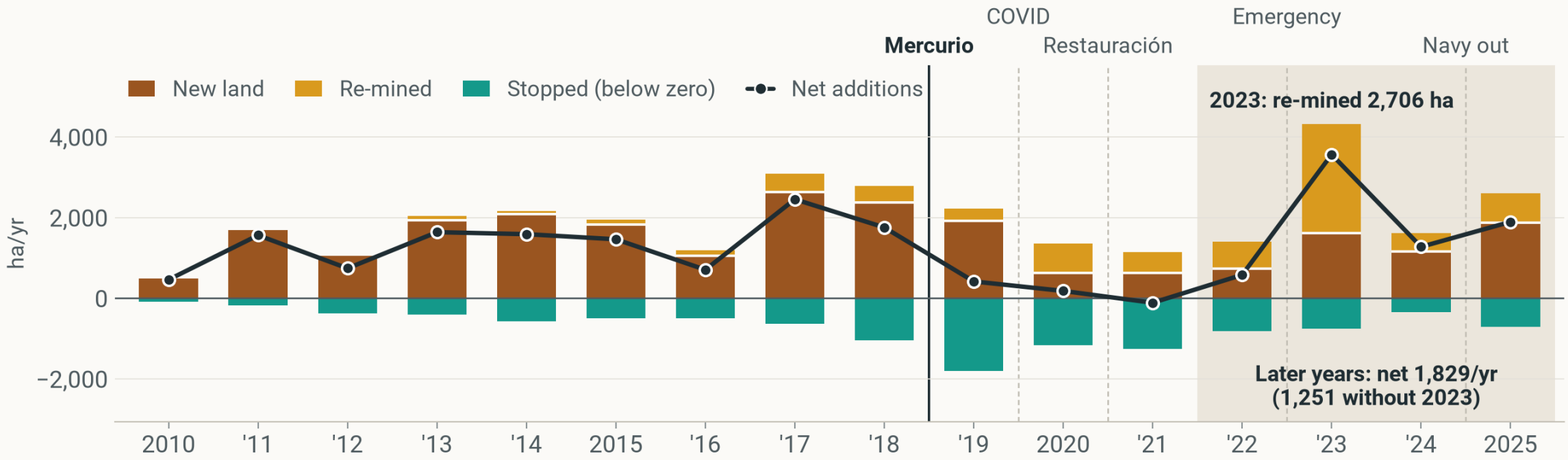


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), our pixel accounting on one zone registry (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; corridor = D.L. 1100). Right: zones inside the department; they sum to the total. Legality is by location; the map class can't tell legal from illegal. Satellites cannot see river dredges or mercury.

DESCRIPTION

It came back on old pits: 46% of added land was re-mined

Net mining additions and their parts, ha/yr · Tambopata buffer zone · 2010–2025 · later years (2022–25): re-mined 1,139/yr, new land 1,345/yr; without the 2023 spike year re-mined 617/yr (33%).

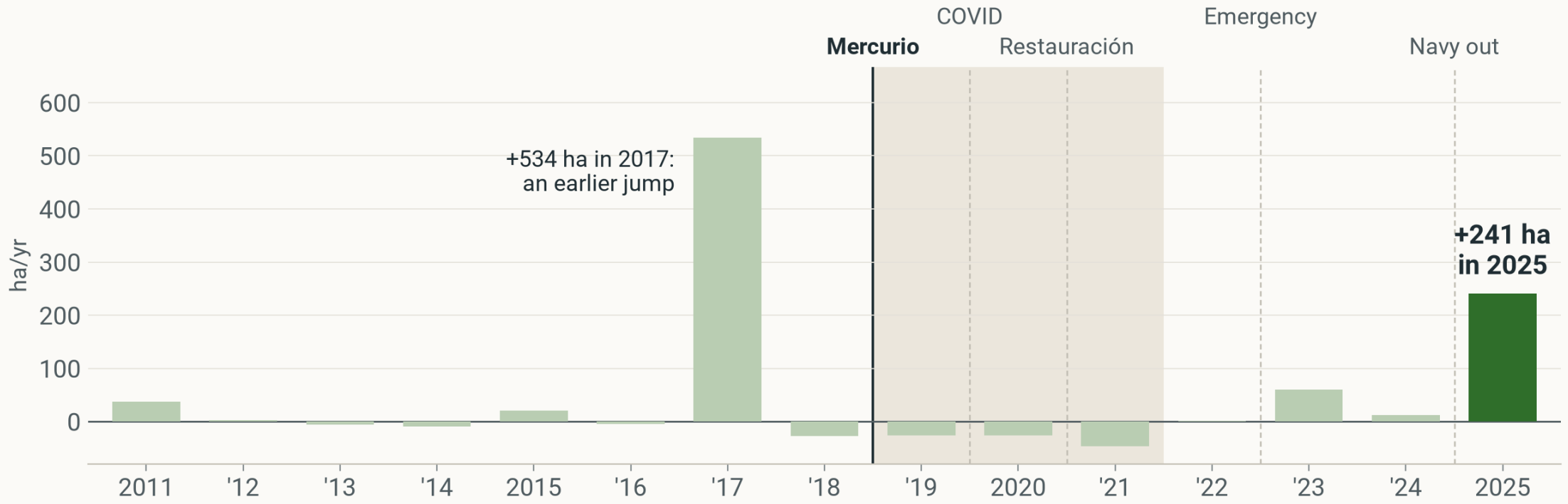


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), our pixel accounting on one zone registry (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; corridor = D.L. 1100). New land = mined for the first time since 1985; re-mined = mined again after a gap; stopped = left the mining class. 2023 is a MapBiomas-only spike (GLAD alerts and Amazon Mining Watch are flat): read 2022–25 with and without it. MAAP #193 (2023) reports miners back on degraded land. Satellites cannot see river dredges or mercury.

DESCRIPTION

2025: mining inside the reserve jumped 241 ha after the navy left

Net mining additions, ha/yr · Tambopata National Reserve (inside the reserve, not the buffer zone) · 2011–2025 · 2019–21 mean −33/yr; mining area 2025: 777 ha. Navy left the Malinowski River posts in 2025 (MAAP #241, 2026).

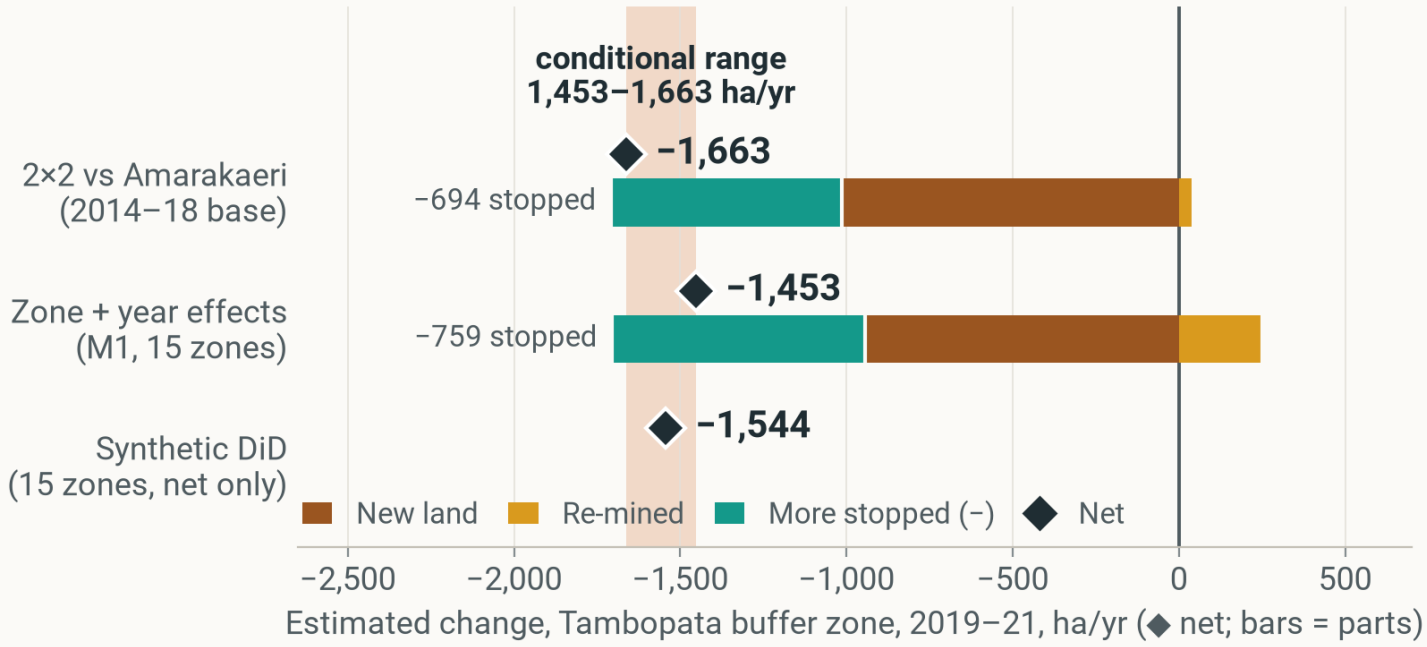


Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), our pixel accounting on one zone registry (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; corridor = D.L. 1100). Platform check: 777 ha of mining inside the reserve in 2025 (protected-area statistics). MAAP #241 (Pacsi et al., 2026) reports ~500 ha inside the reserve, late 2025 to Feb 2026. Satellites cannot see river dredges or mercury.

CAUSAL · CONDITIONAL

Did Mercurio cause the fall? Only conditionally, in hectares only

If the comparison zones show what Tambopata would have done, Mercurio and what followed lowered net additions by 1,453–1,663 ha/yr in 2019–21 (upper bound; part of it pits stopping; coincides with COVID) · Tambopata buffer zone · ha/yr



In hectares, 2019–21 — **1 of 15**

1

In hectares, 2014 (before) — **1 of 15: size**

1

Scaled to own size, 2019–21 — **3–6 of 7**

1

1 = biggest fall. Raw rank 1 also before 2019: it ranks size as much as effect. Not a significance test.

Source: MapBiomas Peru Collection 4 rasters, national buffer-zone panel on the v3 registry (15 zones with pre-2019 mining; scaled ranks: 7 zones above 10 ha/yr). 2×2 = Tambopata minus Amarakaeri, 2019–21 vs 2014–18; M1 = zone + year effects 2014–25; synthetic DiD = own weights (peru_sdid.py, net only). Parts add up to net for the linear estimators.

DESCRIPTION

What satellites cannot see, and which way our biases push

Net mining additions measure land, not people, rivers or mercury · direction of each bias for the 2019–21 fall, Tambopata buffer zone

SATELLITES CAN'T SEE

River dredges

148 → 598 in La Pampa, 2021–23 (MAAP #193)

Mercury

in rivers and fish

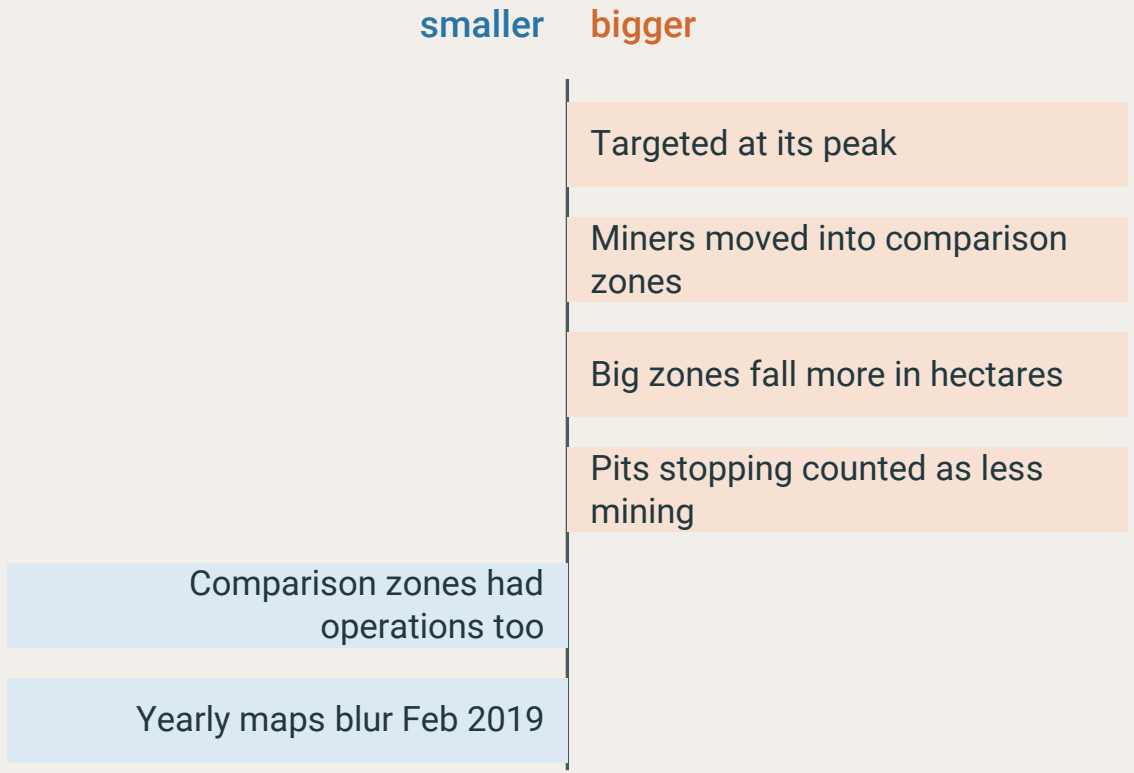
People

hectares, not miners

Map flicker

37% of land that stopped in 2019–21 was mined again within 2 years

EACH BIAS MAKES THE 2019–21 FALL LOOK…



Read the fall as an upper bound, the return as a lower bound.

PREDICTION

Stay after the sweep, and watch the ring around it

Informed expectations from one case (prediction), not proven rules · net mining additions, Tambopata buffer zone, reserve and corridor

Stay after the sweep

163 → 1,829

ha/yr net additions, 2019–21 → later years. Fund presence, not one-off operations.

Watch the ring, in near real time

60 → 395

ha/yr just across the road. Forest-loss alerts on a few-km ring around each operation.

Hold the reserve's first km and the river

91%

of first-time mining inside the reserve lies within 1 km of its edge. Staff the Malinowski posts.

Questions?

Like the anteater, mining comes back for seconds.



Photo: rangers / Tambopata river

For questions

Six attempts · stopped land · road and cells · twins · Brazil · money and gold · definitions and data · map steps

CAUSAL · CONDITIONAL

Six attempts at cause: two hold in hectares only

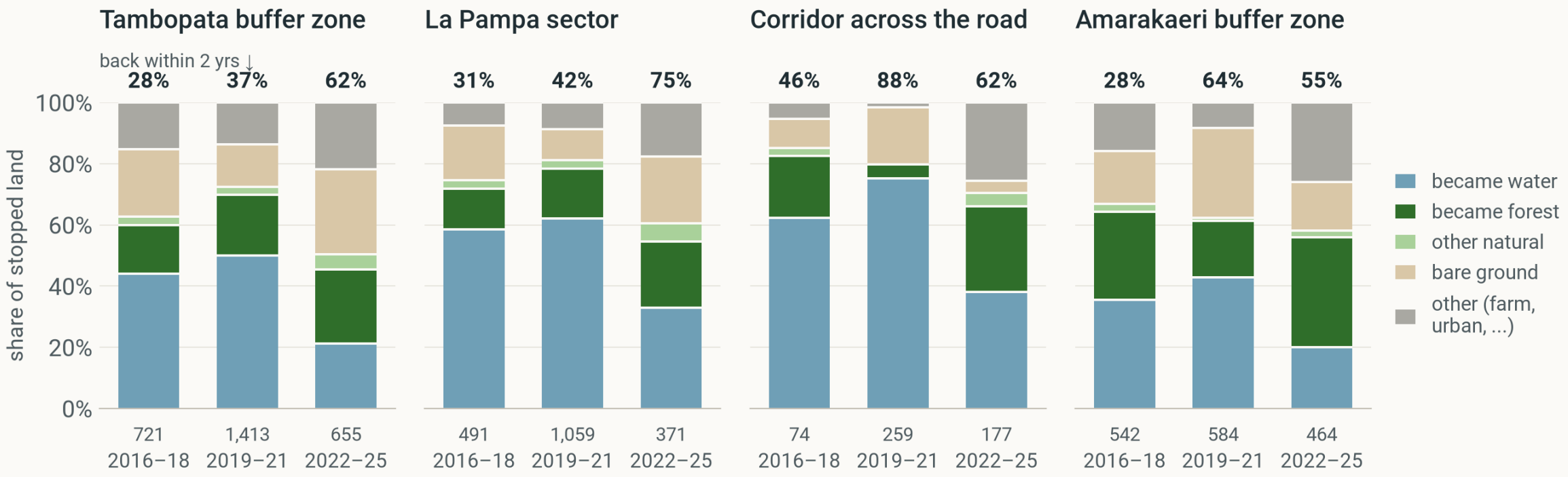
Net mining additions, Tambopata buffer zone, 2019–21 vs before, ha/yr unless stated · all conditional on the comparison

METHOD	COMPARES TAMBOPATA WITH	RESULT (NET)	VERDICT
1 Zone + year effects (M1–M3)	14 Peruvian buffer zones	−1,453 to −1,859; scaled rank 4 of 7	Holds in ha
2 Synthetic control	Weighted mix of zones	−411; no pre-2019 fit in hectares	Weak
3 Rule-based twins	Amanã, Tapajós, Munduruku (Brazil)	16%–41% of their pace (AMW, other measure)	Weak
4 Spatial RD at the road	Either side of the highway	+2.2 pts of land/yr (p = 0.25)	Swing only
5 Matching (7 methods)	Areas picked by algorithm	All slower; placebo p 0.15–0.60	Can't decide
6 Synthetic DiD	Weighted, trend-matched zones	−1,544; scaled rank 3 of 7	Holds in ha

DESCRIPTION

Stopped land: often water, often mined again within 2 years

Stopped land (left the mining class), share by what it became and share mining again within 2 years, period means · four zones · 2016–18, 2019–21, 2022–25 (small numbers under the bars: stopped ha/yr). Numbers on top: share mining again within 2 years.



Source: MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), our pixel accounting on one zone registry (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; corridor = D.L. 1100). Back within 2 years = classifier flicker or re-entry; the last two years cannot be followed up. No image validation sample (plan v3, V3-5).

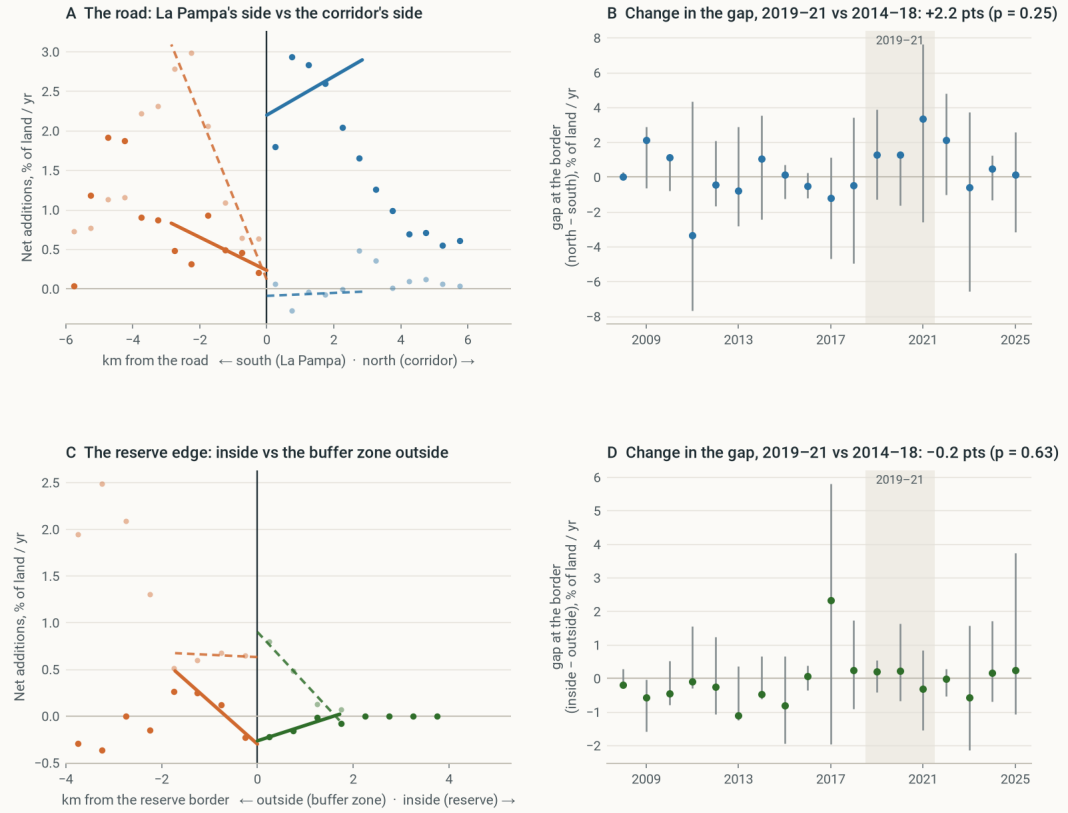
DESCRIPTION

At the road the gap swung north; at the reserve edge nothing

Net mining additions (and parts where linear) · La Pampa longitudes · 2014–25 · see the methods page for the full setup

Road: the gap rose +2.2 pts in 2019–21 (p = 0.25). Reserve edge: no measurable change (−0.2, p = 0.63)

Net mining additions (new + re-mined − stopped), % of land per year, by distance from each border (left), and the gap at the border each year (right). Dashed = 2014–18, solid = 2019–21



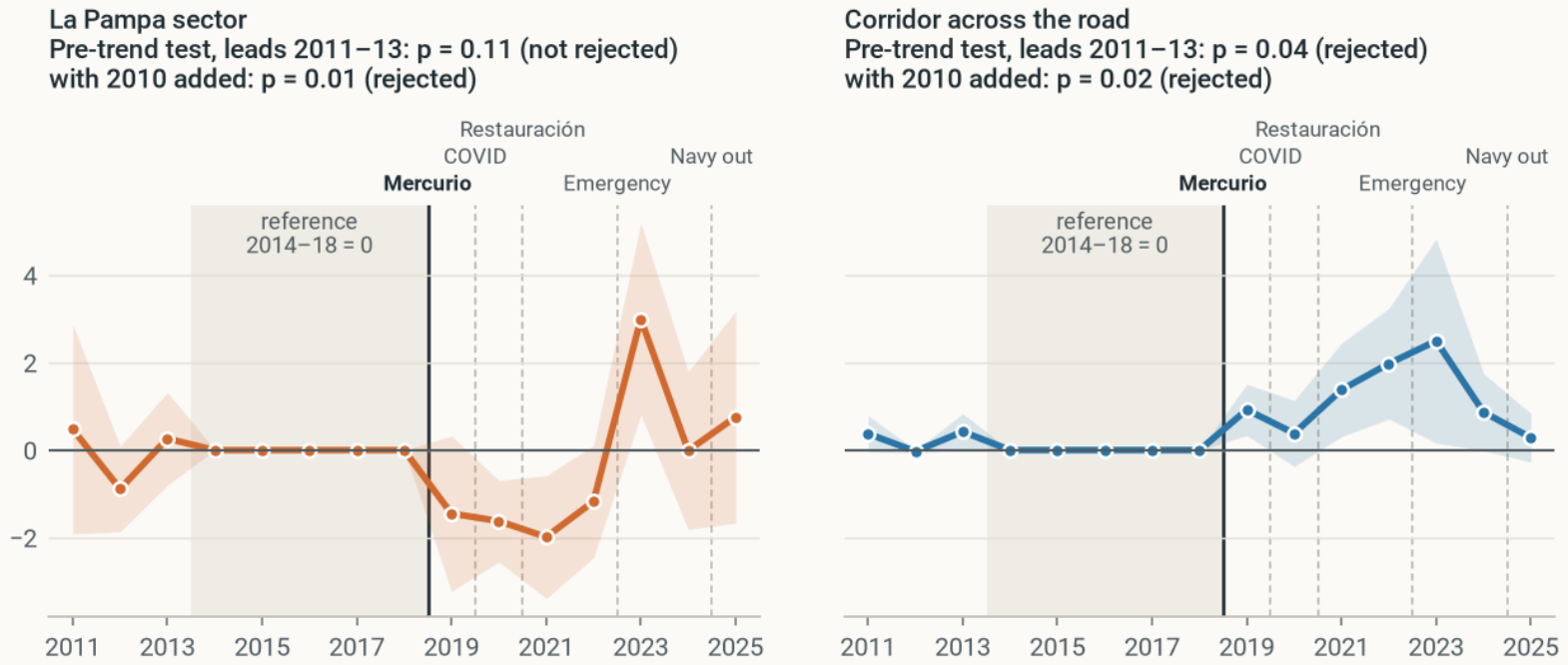
Source: MapBiomas Peru Collection 4 rasters (class 4.2 Minería), our pixel accounting, restreamed by RD unit; SERNANP, RAISG road; rdrobust. Units: 250-m bins in 2-km segments along each border; local linear fits, triangular kernel, bandwidth MSE-optimal for net (road 2.9 km, reserve 1.7 km); 95% robust bias-corrected intervals, CR1 errors clustered by segment. Road: La Pampa longitudes, north-south offset to the road. Description: displacement opens the road gap from both sides, so it measures the swing, not the net effect.

Cell panel: close to a before/after, not a clean comparison

Net mining additions (and parts where linear) · La Pampa longitudes · 2014–25 · see the methods page for the full setup

Net additions per cell: La Pampa clearly below 2014–18 from 2020; across the road clearly above from

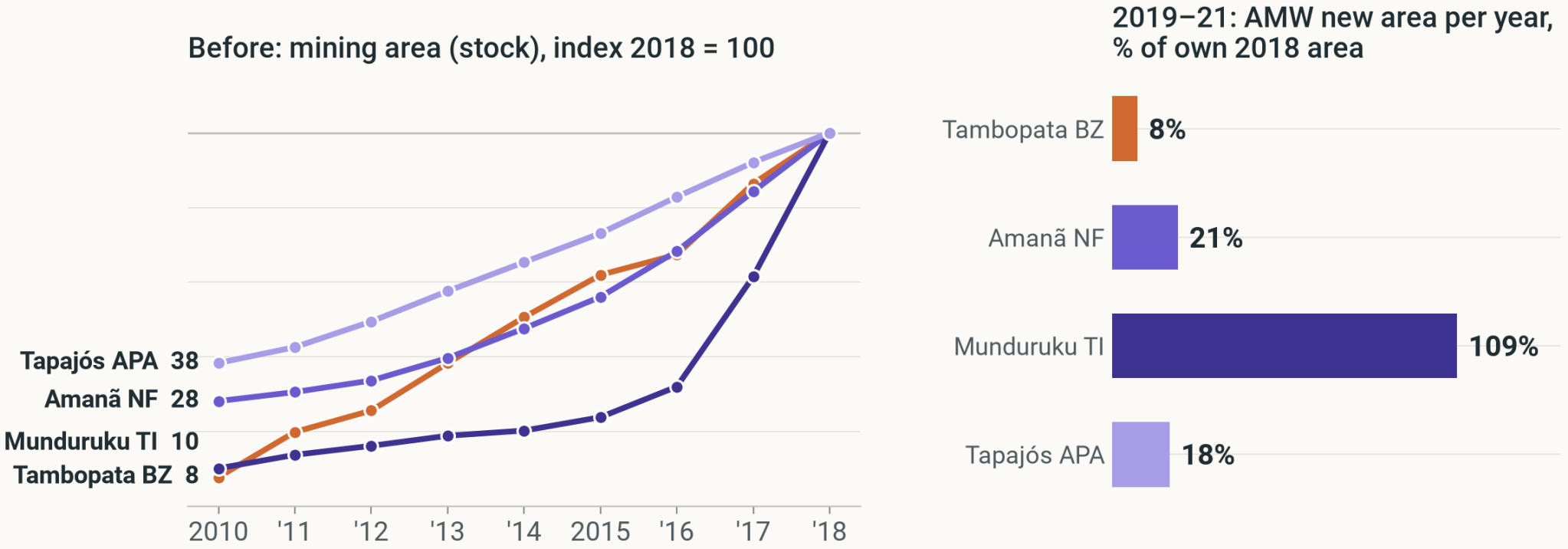
Event study per 1-km cell: net mining additions vs the 2014–18 mean (= 0), ha per cell per year, 95% CI



Source: MapBiomas Peru Collection 4 rasters (class 4.2 Minería), our pixel accounting, 1-km cells (raster_zones.py), majority registry zone; errors clustered by 10-km blocks. Event study with cell and year fixed effects; reference = the pooled 2014–18 period (shaded); comparison cells with little mining (no corridor or community tenure); joint test that the 2011–13 leads are zero. With the 2010 lead added (the v2 window) the test gives p = 0.009 for La Pampa and 0.017 across the road. Description. Vertical lines: an event in year Y is drawn at Y - 0.5, just before the first annual addition that can reflect it (addition(Y) = map(Y) - map(Y-1)).

Twins: Tambopata at 16%–41% of their pace; trends differed before

Left: mining area (stock), index 2018 = 100, 2010–18 (pre-trends rejected, p < 0.001). Right: Amazon Mining Watch new area 2019–21 per year, % of own 2018 area: a different measure, not net additions.



Source: MapBiomas Peru Col 4 rasters, v3 registry (Tambopata) / MapBiomas Amazonía Col 6 (twins); Amazon Mining Watch new area, area-weighted (comparable to first-time mining, not net). Twins chosen by fixed rules (rule R*, control_select.py). Different collections: compared as indices, never subtracted.

DESCRIPTION

Brazil: a boom, then a fall; operations came at the peak

Net mining additions, ha/yr · MapBiomas Brazil (indigenous lands: Collection 11; staggered DiD: Collection 11 rasters, our pixel accounting)

ALL INDIGENOUS LANDS, ARTISANAL

1,332 → 4,319 → 2,978

ha/yr net additions, 2016–18 → 2019–22 → 2023–25

YANOMAMI TERRITORY

500 → 482

ha/yr net additions, 2019–22 → 2023–25 (operation from Feb 2023)

STAGGERED DID, 9 TERRITORIES

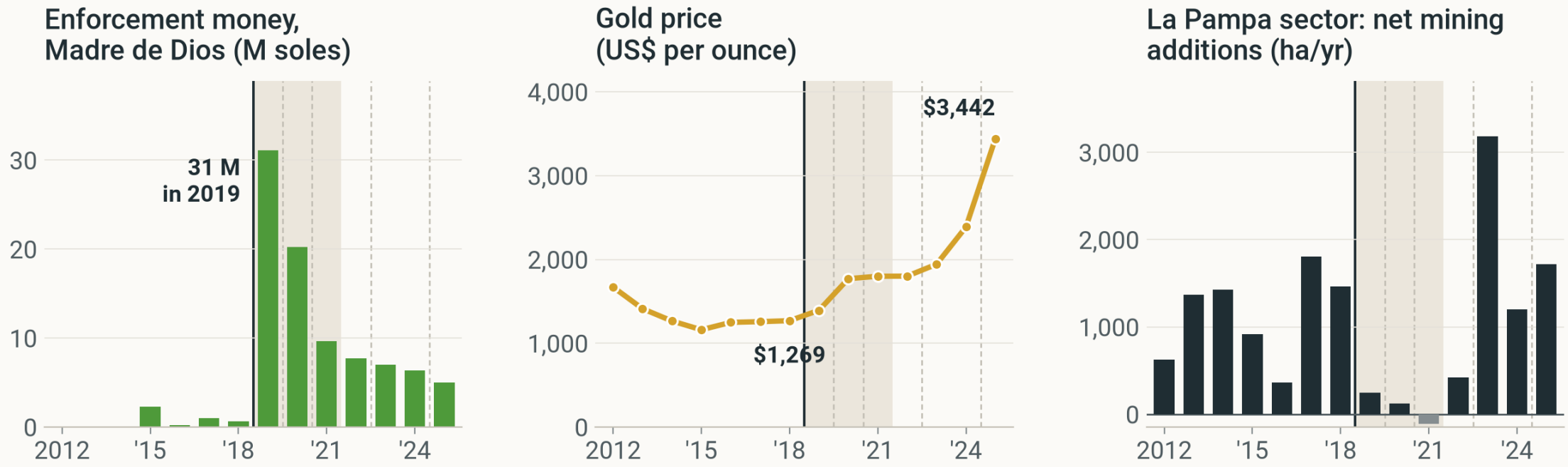
+166

ha/yr net, p = 0.37: cannot separate

Operations came at the peak of a boom, together with national gold-sale reforms (Jan 2023). Our design cannot separate them.

Enforcement money peaked in 2019 and faded; gold rose 2.7-fold

Same years in each panel · PP 0128 executed spending, Madre de Dios (lower bound) · gold price, nominal · net mining additions, La Pampa sector, ha/yr · 2012–2025. Shaded: 2019–21. Context, not a test.



Source: MEF Consulta Amigable, PP 0128 devengado, Madre de Dios; World Bank Pink Sheet (nominal gold price, annual mean); MapBiomas Peru Collection 4 annual maps (30 m, class 4.2 mining, legal and illegal), our pixel accounting on one zone registry (La Pampa sector = Tambopata buffer zone 70.15–69.80° W; corridor = D.L. 1100). Correlation not shown: 11 years, and money follows mining.

DESCRIPTION

Definitions and data

One outcome, one zone registry, one treatment (plan v3) · every number in this deck comes from output/numbers.csv

ITEM	DEFINITION	SOURCE
Net mining additions	Mining area in year t − year t−1, ha/yr = new land + re-mined − stopped	MapBiomas Peru Collection 4 rasters (30 m, class 4.2)
New land / re-mined / stopped	First time since 1985 / mined again after a gap / left the mining class	Our pixel accounting (code/raster_zones.py)
Tambopata buffer zone	Official SERNANP polygon, whole (incl. Kotsimba)	SERNANP
La Pampa sector	Tambopata buffer zone between 70.15° and 69.80° W (no official line)	Our boundary
Corridor across the road	D.L. 1100 corridor ≤ 10 km north of the highway, La Pampa longitudes	INGEMMET, RAISG
Treatment	Operation Mercurio, Feb 2019; first treated year 2019; reference 2018	MAAP #130; MINAM
Periods	2016–18 before · 2019–21 Mercurio and what followed · 2022–25 later years (± 2023)	Course card

DESCRIPTION

The map build as one animation

Same four steps as the click-through map slides · protected areas → highway → corridor → La Pampa · click to play (21 s)

